Ball Trajectory and Arrival Time Prediction System for Shot-Stopping Control of the ERSOW Robot Goalkeeper

1st Dimas Firmansyah
Department of Informatics and Computer Engineering
Politeknik Elektronika Negeri Surabaya
Surabaya, Indonesia
dimasfirmansyah@ds.student.pens.ac.id

2nd Muhammad Nizar Habibi
Department of Electrical Engineering
Politeknik Elektronika Negeri Surabaya
Surabaya, Indonesia
nizar@pens.ac.id

3rd Iwan Kurnianto Wibowo
Department of Informatics and Computer Engineering
Politeknik Elektronika Negeri Surabaya
Surabaya, Indonesia
eone@pens.ac.id

4th Mochamad Mobed Bachtiar
Department of Informatics and Computer Engineering
Politeknik Elektronika Negeri Surabaya
Surabaya, Indonesia
mobed@pens.ac.id

5th Zelcy Wiluarta
Department of Informatics and Computer Engineering
Politeknik Elektronika Negeri Surabaya
Surabaya, Indonesia
zelcy666@ce.student.pens.ac.id

6th Rizki Mia Dwinata
Department of Energy Mechanics Engineering
Politeknik Elektronika Negeri Surabaya
Surabaya, Indonesia
rizkimia@me.student.pens.ac.id

Abstract*— Goalkeeping robots in the RoboCup Middle Size League require accurate prediction of ball trajectory and arrival time to block fast incoming shots. Previous ERSOW research used a CNN-based method that only classified ball direction into right, left, and center, potentially reducing blocking precision. This study proposes a novel end-to-end shot-stopping system that integrates an omnidirectional camera and a RealSense D455 depth camera with a computationally lightweight regression-based machine learning model. Unlike previous approaches that primarily focused on prediction, the proposed system directly applies the predicted interception point and arrival time to the goalkeeper robot's motion control for real-time blocking. By prioritizing RealSense data for long-range ball localization, the system achieved an average positional error below 10 cm in both x and y coordinates. Linear Regression was selected as the optimal model, yielding an MAE of 2.8 cm and a MAPE of 0.62%, while arrival-time prediction produced errors ranging from 0 to 95 ms. Tested at ball speeds of 3–7 m/s, the ERSOW goalkeeper robot successfully blocked all shots, with a maximum deviation of 7 cm from the robot center and an average computation time of 35 ms. These results demonstrate high prediction accuracy and real-time feasibility for effective shot stopping.

***Keywords*— Ball Trajectory Prediction, Goalkeeper Robot, Regression, Camera Integration, RoboCup MSL**

I. Introduction

RoboCup is an annual international robotics competition founded in 1997 with the aim of promoting research in robotics and artificial intelligence by offering an engaging challenge to the public. Its ambitious mission is to build a team of autonomous football robots capable of winning a match against the World Cup winners, according to the official FIFA rules [1]. The Middle Size League (MSL) is one of the divisions in the RoboCup competition that uses autonomous wheeled robots (mobile robots) to play football like humans do [2]. In this competition, the football robots have implemented an autonomous system that requires them to adapt to their surroundings and coordinate with other robots without human intervention [3]. One of the teams competing in the MSL and the KRSBI-B (Kontes Robot Sepak Bola Indonesia Beroda) is ERSOW (EEPIS Robot Soccer on Wheeled) team [4].

In soccer, the team that scores the most goals wins. A goalkeeper must stop shots from opposing players to affect the outcome [5]. Predicting the ball's path is important for robot goalkeepers in RoboCup competitions [6]. Furthermore, according to Article RC-4.2.0, Point 3, when the goal is under threat, the goalkeeper is only allowed to instantly expand the size of their body by a maximum of 10 cm beyond the robot's normal limits (fitting within a 30 cm × 30 cm to 56 cm × 56 cm square), for a maximum duration of 1 second, and must wait a 4-second cooldown period before expanding their body again [7].

In a 2025 study, the EEPIS Robot Soccer on Wheeled (ERSOW) team introduced a Convolutional Neural Network (CNN) algorithm to predict the direction of a ball kick (classifying to categories: right, left, and center) for a robotic goalkeeper [8]. Consequently, the robot may stop the ball using its side area, reducing the effective blocking area and increasing the likelihood of conceding a goal.

To address these limitations, this study proposes a novel end-to-end shot-stopping system for the ERSOW goalkeeper robot by integrating data from an omnidirectional camera [9] and a RealSense D455 depth camera [10]. Unlike the previous CNN-based approach that only classifies the incoming shot into right, left, or center categories [8], the proposed system predicts both the ball interception point and its arrival time using a computationally lightweight regression-based machine learning model. The novelty of this work lies not only in trajectory prediction but also in the direct implementation of the predicted position and arrival time into the goalkeeper robot motion control system, allowing the robot to move and block the ball at the appropriate moment. Furthermore, the predicted arrival time can support the activation of the body-expansion mechanism in accordance with Article RC-4.2.0, Point 3 [7], enabling the robot to maximize its blocking area within the permitted time window. Through this end-to-end integration, the proposed system is expected to improve real-time shot-stopping performance, particularly against fast and varied shots in RoboCup MSL competitions.

II. Related Work

Given the critical role of predicting the ball's trajectory in soccer goalkeeping robots, numerous studies have focused on developing autonomous computational systems to reduce goals conceded. For instance, Setiawan et al. [11] use a trigonometric method based on uniform linear motion, which is computationally lightweight and estimates the ball's

direction from its last and current coordinates, achieving an accuracy of up to 90%.

Apart from the trigonometric method, Setiawardhana et al. [12] applied a more complex computational approach by introducing a modified two-layer neural network (MTLNN) algorithm combined with a fuzzy inference system-based goalkeeper robot navigation (GKRN). While it maintains the same 90% accuracy as the earlier method [11], this approach uniquely predicts the ball's arrival time, enabling more dynamic movement adjustments. Expanding on these advancements, Rahmat et al. [13] employed a Kalman filter and a dual-camera system (omnidirectional and stereo) to filter noise and estimate the ball's velocity, subsequently converting the estimated time to wheel velocity via kinematic inversion. This allowed the robot to achieve 80%–100% interception success at moderate speeds, though performance decreased to 40% at 2 m/s. Similarly, Aditiya et al. [14] used dual cameras (an omnidirectional camera and a front webcam) to collect data for intercepting the ERSOW team's attacking robot. Using a straight-line regression equation, they determined the ball-prediction point and controlled wheel movement via PID, achieving an 80.56% success rate at speeds of 2–4 m/s.

In contrast to MTLNN [12], Kalman Filter [13], and straight-line regression equation [14], Yano et al. [15] used Long Short-Term Memory (LSTM) to analyze time series of ball movements by considering the temporal dependency between consecutive positions, resulting in Mean Squared Error (MSE) values below 0.1. However, neural network-based approaches generally have a relatively high computational burden. Thus, there is a need for a system that can acquire ball position data and make real-time predictions with minimal computational cost to improve the performance of the goalkeeper robot.

III. Proposed Method

Fig. 1 shows the workflow diagram of the team ERSOW goalkeeper robot system, which consists of image processing, ball trajectory prediction, robot localization, and actuator control modules. The computer vision system integrates an omnidirectional camera [9] and a RealSense D455 depth camera [10] to detect the ball position and distance. The detected ball data are then processed by a lightweight regression-based machine learning model to estimate both the interception point and arrival time. Unlike prediction-only approaches, the proposed method directly uses these predictions as inputs to the goalkeeper robot's motion control system, enabling real-time shot-stopping. The entire image processing and prediction system runs on an ASUS Mini PC equipped with an AMD Ryzen 5 5500U processor and AMD Radeon Graphics to support real-time computation. The predicted position and arrival time are forwarded to the intelligent control system before being sent to the mainboard. The robot's movement is controlled using a PID controller to achieve the required speed and track the predicted target position [14]. The robot position on the field is estimated using a combination of an inertial measurement unit (IMU) and odometry sensors [16], enabling it to calculate its deviation from the target.

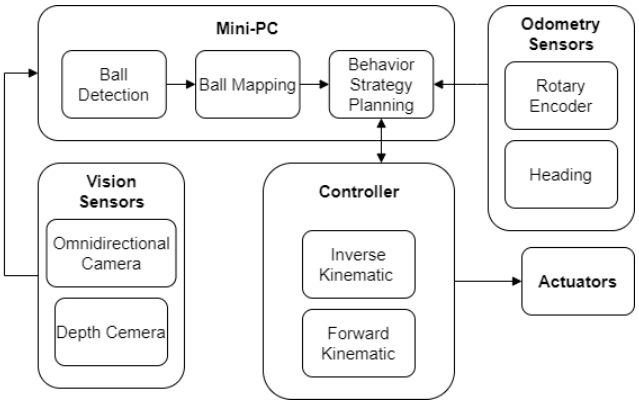


Fig. 1. ERSOW Team Goalkeeper Robot Workflow

A. Ball Detection Method

To improve detection accuracy, the ERSOW goalkeeper robot integrates data from two different cameras. This integration is necessary because omnidirectional cameras lose resolution significantly as objects become farther away [17], whereas depth cameras such as the RealSense D455 can detect objects farther away with low error, namely 191.4 mm (X), 123.47 mm (Y), and 222.15 mm (Z)[10]. The placement of the two cameras on the robot structure is shown in Fig. 2.

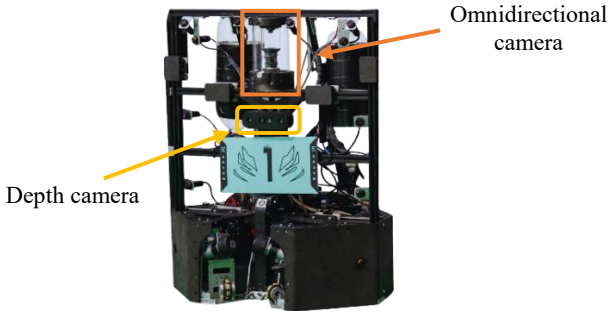
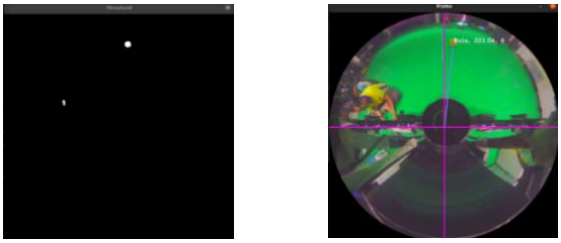


Fig. 2. ERSOW Team Goalkeeper Robot

One popular approach to this detection is color segmentation, where the image is converted to a specific color space to facilitate object identification. HSV representation is one of the color spaces shown to be more stable against lighting variations and changes in image size, making this approach effective for applications across various real-image conditions [18]. In the ERSOW team, the color thresholding method in the HSV color space was applied to an omnidirectional camera [9] and a depth camera to detect the ball's position in real time. This algorithm works by finding the largest contour based on the color thresholding illustrated in Fig. 3.



(a) Thresholding Result (b) Ball Tracking Result

Fig. 3. Ball Detection in Omnidirectional Camera

After the ball can be detected, the next step is to convert the distance from pixels to centimeters (real-world) using the

regression method [19]. The ball position is represented in the global field coordinate system, where (x_{robot}, y_{robot}) denotes the robot position and $(x_{ball_omni}, y_{ball_omni})$ denotes the estimated ball position. These coordinates are obtained from the measured ball distance (D_{ball_omni}) and orientation (θ_{ball_omni})relative to the robot, as described in Equations (1) and (2).

$$x_{ball_omni} = x_{robot} + D_{ball_omni} sin\theta_{ball_omni} \quad (1)$$

$$y_{ball_omni} = y_{robot} + D_{ball_omni} cos\theta_{ball_omni} \quad (2)$$

Meanwhile, for ball detection on the D455 RealSense camera, the detection system still uses a thresholding method based on HSV color segmentation, as shown in Fig. 4 (a) and (b). The distance of the robot to the ball is measured using a built-in library, get_distance() [10], which calculates a distance (depth) for each pixel along with its RGB intensity, allowing 3D coordinate projection [20] as shown in Fig. 4 (c). The global ball position is represented by $(x_{ball_realsense}, y_{ball_realsense})$. The camera offset $(x_{cam_offset}, y_{cam_offset})$ corresponds to the x- and y-components of the fixed 11 cm displacement between the RealSense camera and the robot center, computed according to the robot heading (ω_{robot}). The global ball coordinates are then calculated from the measured ball distance $(D_{ball_realsense})$ and the ball orientation relative to the robot $(\theta_{ball_realsense})$, as expressed in Equations (3) and (4).

$$x_{ball_realsense} = x_{robot} + x_{cam_offset} + D_{ball_realsense} \cdot \sin(\theta_{ball_realsense} - \omega_{robot}) \quad (3)$$

$$y_{ball_realsense} = y_{robot} + y_{cam_offset} + D_{ball_realsense} \cdot \cos(\theta_{ball_realsense} - \omega_{robot}) \quad (4)$$



(a) Thresholding Result (b) Ball Tracking Result (c) Depth Projection

Fig. 4. Ball Detection in Depth Camera

The dual-camera integration prioritizes data generated by the depth camera because it provides more stable long-range positioning accuracy. However, if the depth camera fails to detect the ball, the ball position is retrieved from the omnidirectional camera.

B. Ball Trajectory Prediction

The quality of predictions is highly dependent on data collection and preprocessing [21], as these steps affect the validity of the ball trajectory data. To capture the unique motion pattern of each shot, data collection and modelling are performed independently for every new kick, allowing the regression model to adapt to variations in kick direction and speed. Previous data are cleared before recording the ball's initial position. If the ball moves back more than 100 cm, the motion is classified as dribbling and the data are discarded. During a valid shot, ball positions are recorded at 8 cm intervals and forwarded to preprocessing once the ball has traveled more than 150 cm and reaches a Y position below 300 cm. The prediction workflow is shown in Fig. 5.

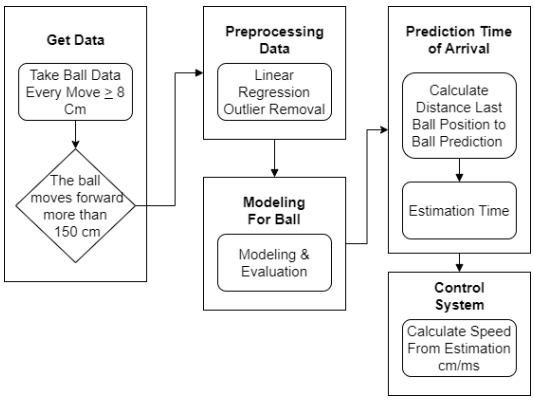


Fig. 5. Ball Trajectory Prediction Workflow

After the dataset is collected, data cleaning is performed by examining each trajectory point in reverse order. For every candidate point, a linear regression line is fitted using the subsequent trajectory samples, and the perpendicular distance d from the point (x_i, y_i) to the fitted line is calculated using Equation (5), where m and c are the slope and intercept estimated from the subsequent trajectory samples using the least-squares regression method. Points whose distances exceed the predefined outlier threshold are considered outliers and removed before the prediction stage.

$$d = \frac{|x_i - (m \cdot y_i + c)|}{\sqrt{1+m^2}} \quad (5)$$

The results of outlier removal are shown in Fig. 6, where each outlier caused by camera noise is removed as much as possible.

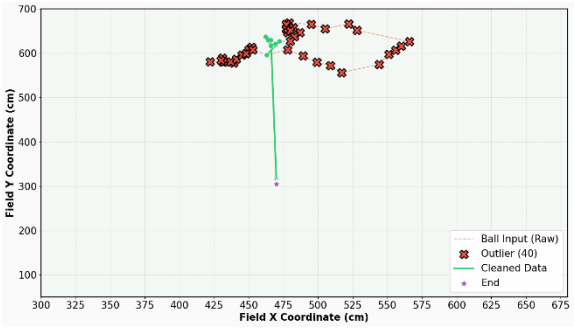


Fig. 6. Outliers Removal Result

To improve the quality of the analysis, the cleaned data were further processed by adding distance-feature information $(\Delta x_{bola}, \Delta y_{bola}, \Delta t_{bola})$ to convert the static position data into a velocity vector representation, enabling the robot to estimate the ball direction and speed.

After the ball data were converted to a velocity vector, the next step was to build a ball position prediction model with the x_{ball} variable as the target. Model inputs included the previous x_{ball} and y_{ball} positions, as well as changes in Δx_{ball}, Δt_{bola},and Δy_{ball} which allowed the goalkeeper robot to capture the ball's motion patterns. Linear Regression was used as a baseline to predict the next x_{ball} position from the previous ball position and motion features, as formulated in Equation (6), where β_0 is the intercept, while $\beta_1 x_t$, $\beta_2 y_t$, $\beta_3 \Delta x_t$, and $\beta_4 \Delta y_t$ represent the weighted contributions of the

previous ball position and displacement to the predicted position, and ε denotes the residual error term.

$$\hat{x}_{t+1} = \beta_0 + \beta_1 x_t + \beta_2 y_t + \beta_3 \Delta x_t + \beta_4 \Delta y_t + \varepsilon \quad (6)$$

Polynomial Regression extends the Linear Regression model by augmenting the original input features with second-order polynomial terms, enabling the model to capture non-linear relationships between the ball position and its motion. In addition to the linear features, the second-degree polynomial introduces the quadratic terms x_t^2 and y_t^2, which allow the influence of the ball position to vary non-linearly, as well as the interaction term $x_t y_t$, which models the combined effect of the horizontal and vertical positions on the predicted trajectory. The polynomial degree was set to two to increase the model's representational capability while maintaining relatively low computational complexity. The resulting prediction model is formulated in (7).

$$\hat{x}_{t+1} = \beta_0 + \beta_1 x_t + \beta_2 y_t + \beta_3 x_t^2 + \beta_4 y_t^2 + \beta_5 x_t y_t + \beta_6 \Delta x_t + \beta_7 \Delta y_t + \varepsilon \quad (7)$$

Meanwhile, Ridge Regression extends the Linear Regression model by introducing an L2 regularization term to the objective function, where the L2 norm represents the sum of the squared regression coefficients. This regularization reduces the sensitivity of the regression coefficients to correlated input features, resulting in a more stable solution when the ball position and motion features exhibit multicollinearity. The regularization strength was set to $\alpha = 10$, providing a moderate penalty that limits coefficient magnitudes without excessively shrinking their contributions. The optimization objective for predicting the next ball position $\hat{x}_{t+1}$ is described in Equation (8), where β_0 is the intercept, β_1–β_4 are the regression coefficients learned from the training data, x_i denotes the observed target value of the i-th training sample, and n is the total number of training samples.

$$\hat{x}_{t+1} = arg\min_{\beta} \sum_{i=1}^{n} (x_i - (\beta_0 + \beta_1 x_t + \beta_2 y_t + \beta_3 \Delta x_t + \beta_4 \Delta y_t))^2 + 10 \sum_{j=1}^{4} \beta_j^2 \quad (8)$$

Once the prediction of x_{ball} is obtained, the prediction of y_{ball} is performed similarly by using the predicted x_{ball} value as an additional input. The ball travel time (Δt) is then estimated using a Linear Regression model based on the displacement features $(\Delta x_t, \Delta y_t)$ and the previous travel time (Δt_t), as described in Equation (9), where α_0 is the intercept, α_1–α_3 are the regression coefficients learned from the training data, and ε denotes the residual error term.

$$\widehat{\Delta t}_{t+1} = \alpha_0 + \alpha_1 \Delta x_t + \alpha_2 \Delta y_t + \alpha_3 \Delta t_t + \varepsilon \quad (9)$$

A separate regression model is built from each kick's trajectory samples and used to extrapolate the ball's x-coordinate at the target line (y = 30 cm), beyond the observed segment. The resulting out-of-sample errors are evaluated over 10 independent kicks (Table II), with MAE (Mean Absolute Error), and MAPE (Mean Absolute Percentage Error) computed relative to the ground-truth intercept from the last camera-detected ball position.

C. Motion Control System

The robot control system is designed to execute precise movements based on the predicted ball trajectory. To achieve omnidirectional motion, the robot employs inverse kinematics to transform the desired robot velocity into the rotational speed of each wheel motor. As illustrated in Fig. 7, the robot consists of four omni wheels arranged asymmetrically. The desired robot motion is represented by the translational velocities V_x and V_y along the robot's x and y-axes, and the rotational velocity V_ω. These motion commands are converted into the individual wheel velocities v_1–v_4 using the inverse kinematics relationship described in Equation (10), where L_1 and L_2 denote the distances from the robot center to the front and rear wheel pairs, respectively.

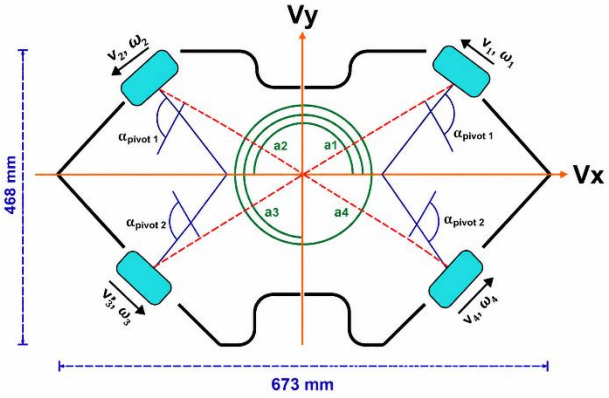


Fig. 7. Omni Four Wheel Position Design

$$\begin{bmatrix} v_1(t) \\ v_2(t) \\ v_3(t) \\ v_4(t) \end{bmatrix} = \begin{bmatrix} -\sin(47.5°) & \cos(47.5°) & L_1 \cdot \cos(10.76°) \\ -\sin(132.5°) & \cos(132.5°) & L_1 \cdot \cos(10.76°) \\ -\sin(227.5°) & \cos(227.5°) & L_2 \cdot \cos(14.23°) \\ -\sin(312.5°) & \cos(312.5°) & L_2 \cdot \cos(14.23°) \end{bmatrix} \begin{bmatrix} v_x(t) \\ v_y(t) \\ v_\omega(t) \end{bmatrix} \quad (10)$$

For robot localization, the system estimates the robot position using encoder and IMU data. Two encoder wheels are arranged in a V-shaped configuration to measure the robot's motion while minimizing the influence of wheel slippage from the main drive wheels. The robot displacement in the local coordinate frame is represented by Δx_{local} and Δy_{local}, while v_1 and v_2 denote the displacement measurements obtained from the two encoder wheels. The heading angle θ, estimated from the IMU, is used to rotate the encoder measurements into the robot's local coordinate frame, as described in Equation (11).

$$\begin{bmatrix} \Delta x_local \\ \Delta y_local \end{bmatrix} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \quad (11)$$

The main input to the control system is the required X-axis robot speed (V_x), which is calculated based on the predicted ball trajectory. This value is then converted using a speed-search calibration to map motor control to the robot actual speed, which is not perfectly linear due to wheel slip, as shown in Fig. 8. Next, the Y-axis (V_y) and the robot rotation angle (V_t) are stabilized using a PID controller, enabling the robot to precisely adjust its lateral movement and orientation.

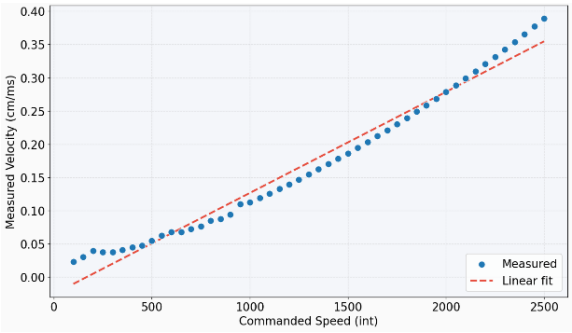


Fig. 8. Robot Speed Lookup Visualization

IV. Results And Discussion

This section presents the experimental results of the proposed shot-stopping system. The same ten shooting trials were used throughout all experiments, while the results are discussed separately according to each stage of the system. To emulate realistic match conditions, the ball was kicked from randomly varied initial positions and directions, since shots from opposing robots may originate from different field locations. The kicking speed was varied between 3 and 7 m/s to represent fast shots commonly encountered during RoboCup matches and those generated by the ERSOW team attacking robot. Ball detection was evaluated independently, whereas the trajectory prediction and shot-stopping experiments shared the same shooting conditions, including the initial ball positions and kicking angles listed in Table I.

TABLE I. Ball Initial Positions and Shot Directions

First Ball Location (cm)		Kick Angle Abs (°)
X	*Y*	
274	416	67
369	704	92
370	672	93
408	716	83
26	417	41
242	593	67
380	658	82
151	707	66
611	700	106
508	714	102

A. Ball Detection

This test was conducted to evaluate the ball detection performance of an omnidirectional camera and a RealSense front camera against the actual ball position on the field. The goal of this test was to compare the accuracy and stability of the two cameras in detecting the ball's coordinates in real time.

TABLE II. Ball Detection Results

Actual (cm)	Omnidirectional Cam (cm)		RealSense Front Cam (cm)	
Ball Position (x, y)	*Ball Position (x, y)*	*Error (x, y)*	*Ball Position (x, y)*	*Error (x, y)*
400, 30	400, 33	0, 3	undetected	-
400, 90	400, 88	0, 2	400, 91	0, 1
300, 400	313, 372	13, 28	294, 401	6, 1
500, 400	495, 390	5, 10	498, 405	2, 5
300, 500	315, 455	15, 45	294, 507	6, 7
600, 500	547, 397	53, 103	592, 508	8, 8
200, 600	244, 490	44, 110	193, 608	7, 8
400, 600	373, 541	27, 59	393, 607	7, 7
500, 600	480, 531	20, 19	492, 607	2, 7
400, 800	undetected	-	401, 808	1, 8

Based on comparisons between ball mapping and omnidirectional cameras, the cameras accurately detected the ball when it was close. However, its accuracy decreased with increasing distance, resulting in errors exceeding 100 cm in some tests beyond 500 cm and occasionally failing to detect the ball due to reduced object resolution. In contrast, the RealSense cameras produced more consistent results, maintaining an average error below 10 cm in both the x- and y-axes. These results indicate that depth-based sensing provides more reliable long-range ball detection than regression-based estimation, supporting the prioritization of RealSense data in the proposed dual-camera system.

B. Ball Trajectory Prediction

To evaluate the trajectory prediction system's performance, the actual and predicted ball positions were compared. In this study, the actual ball position refers to the last ball position detected by the camera, whereas the predicted ball position refers to the estimated ball position when the ball reaches the target line at y = 30 cm. Table III presents the prediction results obtained using Linear, Ridge, and Polynomial Regression, while Table IV evaluates their accuracy using MAE and MAPE derived from the prediction errors shown in Table III. Furthermore, Table V presents the performance of the selected model in predicting the ball arrival time for goalkeeper synchronization.

TABLE III. Ball Trajectory Prediction Results

Actual	Linear Regression		Ridge Regression		Polynomial Regression	
Last Ball Position (x, y)	*Ball Position (x, y)*	*Error (x)*	*Ball Position (x, y)*	*Error (x)*	*Ball Position (x, y)*	*Error (x)*
437, 31	437, 30	0	437, 30	0	432, 30	5
340, 25	340, 30	0	339, 30	1	336, 30	4
341, 46	339, 30	2	339, 30	2	367, 30	26
488, 45	481, 30	7	481, 30	7	488, 30	0
451, 42	452, 30	1	451, 30	0	147, 30	304
484, 32	490, 30	6	502, 30	18	455, 30	29
457, 51	451, 30	6	451, 30	6	425, 30	32
443, 30	441, 30	2	441, 30	2	444, 30	1
415, 31	412, 30	3	412, 30	3	250, 30	165
364, 56	363, 30	1	360, 30	4	417, 30	53

The comparison results indicate that Linear Regression provides the most accurate and consistent trajectory predictions among the evaluated methods. As shown in Tables III and IV, its prediction errors remained relatively small across all tests, with a maximum error of 7 cm, an MAE of 2.8 cm, and a MAPE of 0.62%. Ridge Regression achieved comparable performance but exhibited larger deviations in several trials, resulting in an MAE of 4.3 cm and a MAPE of 0.96%, likely due to the regularization term reducing sensitivity to changes in ball direction and velocity. In contrast, Polynomial Regression produced unstable predictions with errors reaching 304 cm, leading to substantially higher MAE and MAPE values of 61.9 cm and 14.49%, respectively, indicating a tendency to overfit the trajectory data. Therefore, Linear Regression was selected for the subsequent stage of ball arrival time prediction.

TABLE IV. Ball Trajectory Prediction Evaluation

Method	MAE (cm)	MAPE (%)
Linear Regression	2.8	0.62%
Ridge Regression	4.3	0.96%
Polynomial Regression	61.9	14.49%

Table V then presents a test to evaluate the system's ability to predict the ball's arrival time (Δt) relative to the actual value, supporting the goalkeeper robot's movement synchronization.

TABLE V. BALL ARRIVAL TIME PREDICTION RESULTS

Actual		Prediction	
Δt (ms)	*Last Observed Data (x, y)*	*Δt (ms)*	*Error (ms)*
840	344, 250	840	0
573	351, 313	573	0
550	360, 306	550	0
877	451, 311	801	-76
1142	141, 303	1237	95
954	370, 306	1016	62
408	425, 306	319	-86
709	305, 347	742	36
568	490, 296	501	-67
472	409,312	383	-89

Based on the comparison of actual and predicted times in Table V, all recorded arrival time deviations were below 100 ms, and three predictions yielded zero error. This indicates that the proposed method provides sufficient temporal accuracy for real-time goalkeeper movement planning and shot-saving execution. The remaining prediction errors are primarily due to error propagation from the trajectory prediction stage, as the estimated arrival time is calculated based on the predicted ball position, whereas the actual arrival time is determined from the camera-detected ball position. This difference in reference position causes deviations in the estimated arrival time. However, the resulting deviation is still within the goalkeeper robot's response tolerance limits.

This prediction system includes ball position estimation and arrival time, which requires an average computation time of 35 ms on an AMD Ryzen 5 5500U processor with AMD Radeon graphics, demonstrating sufficient computational efficiency for real-time applications.

C. Shot Stopping Motion Control

To evaluate overall system performance, the goalkeeper robot was tested through shot-stopping experiments using predicted ball trajectory and arrival time as inputs to the motion control system. Table VI summarizes performance under various kicking conditions and the impact of prediction accuracy on movement response and shot-stopping success. Robot movement speed (in cm/ms) was calculated by dividing the distance between the robot and the predicted interception point by the predicted ball arrival time. The results show that the robot's speed depends on both the ball's speed and the distance to the predicted position, both of which are related.

TABLE VI. SHOT STOPPING RESULTS

Ball Speed	Motion Robot				Condition
	Distance Ball With Center Robot	*Error To Target*	*Speed (abs)*	*cm/ms*	
3-4 m/s	0	0	400	0.044	Blocked
4-5 m/s	0	0	950	0.1047	Blocked
	-2	0	1000	0.1109	Blocked
	-7	0	950	0.1011	Blocked
5-6 m/s	-3	-4	400	0.042	Blocked
	0	-6	900	0.0881	Blocked
	-6	0	1050	0.159	Blocked
6-7 m/s	-4	-2	550	0.055	Blocked
	3	0	150	0.0239	Blocked
	-1	0	950	0.0967	Blocked

Based on the test, the robot successfully blocked in all scenarios with ball speed variations of 3–7 m/s, with a deviation from the robot center ranging from 3 to 7 cm and a maximum error to the target of 6 cm, which indicates that the robot moved quite close to the predicted intercept point. These results indicate that the trajectory prediction and motion control system is still able to maintain the accuracy of the robot's movement even though there is an error in the prediction of the ball's arrival time of up to tens of milliseconds, because the robot dimensions (67 cm wide) still provide an effective blocking area with a deviation tolerance of up to ±27 cm.

V. CONCLUSIONS

This study presents a novel end-to-end shot-stopping system that integrates an omnidirectional camera and a RealSense D455 depth camera with a lightweight regression-based machine learning model to predict ball trajectory and arrival time for the ERSOW goalkeeper robot. The main contribution of this work lies in directly applying the predicted interception point and arrival time to the robot motion control system, rather than limiting the prediction output to offline trajectory estimation. In the data integration process, the RealSense camera produced more stable measurements, with an average error below 10 cm, making it the primary sensor in the fusion scheme. Linear Regression achieved the best trajectory prediction performance, with an MAE of 2.8 cm and a MAPE of 0.62%, outperforming Ridge and Polynomial Regression, which tended to overfit. The arrival-time prediction also showed acceptable temporal accuracy, with errors ranging from 0 to 95 ms. Experimental results at ball speeds of 3–7 m/s, representing realistic RoboCup shooting conditions and the kicking performance of the ERSOW attacking robot, confirmed that the system successfully blocked all shots, with a maximum deviation of 7 cm from the robot center. Overall, the system performed efficiently with an average computation time of 35 ms, making it feasible for real-time implementation.

ACKNOWLEDGMENT

The authors acknowledge financial support from the 2026 Internal Research Grant of Politeknik Elektronika Negeri Surabaya (PENS), Contract No. 012/SP2H/PENS/PPK/2026.

REFERENCES

[1] W. Houtman, C. A. Lopez Martinez, S. Wang, A. Ketels, H. P. J. Bruyninckx, and M. J. G. van de Molengraft, "Dynamic control of steerable wheeled mobile platforms applied to an eight-wheeled RoboCup Middle Size League soccer robot," *Mechatronics*, vol. 80, p. 102693, Dec. 2021, doi: 10.1016/j.mechatronics.2021.102693.

[2] M. A. Ismail, D. Purwanto, and A. Arifin, "Soccer Robot Localization Based on Sensor Fusion From Odometry and Omnivision," in *2022 International Seminar on Intelligent Technology and Its Applications (ISITIA)*, IEEE, Jul. 2022, pp. 273–278. doi: 10.1109/ISITIA56226.2022.9855313.

[3] A. F. A. Ribeiro, A. C. C. Lopes, T. A. Ribeiro, N. S. S. M. Pereira, G. T. Lopes, and A. F. M. Ribeiro, "Probability-Based Strategy for a Football Multi-Agent Autonomous Robot System," *Robotics*, vol. 13, no. 1, p. 5, Dec. 2023, doi: 10.3390/robotics13010005.

[4] S. B. Jumantoro, I. K. Wibowo, M. M. Bachtiar, F. Ardila, D. M. Sari, and R. Aditiya, “Omnidirectional Vision-Based Localization for ERSOW Middle Size League Soccer Robot,” in *2025 International Electronics Symposium (IES)*, IEEE, Aug. 2025, pp. 376–382. doi: 10.1109/IES67184.2025.11161811.

[5] N. Numazu, Y. Hirashima, and K. Matsukura, “Analysis of soccer goalkeeper performance and shot scenarios in the 2022 World Cup,” *Journal of Physical Education and Sport*, vol. 24, no. 12, pp. 2115–2125, Dec. 2024, doi: 10.7752/jpes.2024.12310.

[6] Y. Mirmohammad, S. Khorsandi, M. N. Shahsavari, B. Yazdankhoo, and S. Sadeghnejad, “Ball Path Prediction for Humanoid Robots: Combination of k-NN Regression and Autoregression Methods,” in *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, vol. 13132 LNAI, Springer Science and Business Media Deutschland GmbH, 2022, pp. 3–14. doi: 10.1007/978-3-030-98682-7_1.

[7] Robocup, “Middle Size Robot League Rules and Regulations for 2025,” 2024.

[8] Setiawardhana, I. Kurnianto Wibowo, and N. Achmad Husein Bernardt, “Prediction of Ball Position Using CNN Methods With Zed Camera on Goalkeeper Robot Application,” *IEEE Access*, vol. 13, pp. 41559–41570, 2025, doi: 10.1109/ACCESS.2025.3546346.

[9] M. A. Haq, I. K. Wibowo, and B. S. B. Dewantara, “Improving the speed of ball detection process and obstacle detection process in ERSOW robot using omnidirectional vision based on ROS,” *Indonesian Journal of Electrical Engineering and Computer Science*, vol. 22, no. 3, p. 1365, Jun. 2021, doi: 10.11591/ijeecs.v22.i3.pp1365-1371.

[10] D. P. Suwandi, I. K. Wibowo, and M. M. Bachtiar, “Estimation of Ball Position Using Depth Camera for Middle Size Goalkeeper Robot,” in *2022 International Electronics Symposium (IES)*, IEEE, Aug. 2022, pp. 374–379. doi: 10.1109/IES55876.2022.9888575.

[11] D. Bagus Setiawan *et al.*, “Ball Direction Prediction for Wheeled Soccer Robot Goalkeeper Using Trigonometry Technique,” *Applied Technology and Computing Science Journal*, vol. 2, no. 1, pp. 39–51, Sep. 2019, doi: 10.33086/atcsj.v2i1.1204.

[12] Setiawardhana, Rudy Dikairono, Djoko Purwanto, and Tri Arief Sardjono, “Navigasi Robot Penjaga Gawang Berdasarkan Prediksi Posisi dan Waktu Kedatangan Bola,” *Jurnal Nasional Teknik Elektro dan Teknologi Informasi*, vol. 9, no. 3, pp. 296–304, Aug. 2020, doi: 10.22146/.v9i3.295.

[13] F. Rahmat, A. Wibisana, M. R. Fauzi, M. Aliman, and N. Firmansyah, “Implementation of Intercept Ball Algorithm for Wheeled Soccer Robot Barelang63,” in *Proceedings of the 7th International Conference on Applied Engineering (ICAE 2024)*, vol. 251, Atlantis Press / Springer Nature, 2024, pp. 190–205. doi: 10.2991/978-94-6463-620-8_15.

[14] R. Aditiya, I. K. Wibowo, M. M. Bachtiar, F. Ardilla, D. M. Sari, and S. B. Jumantoro, “Optimized Ball Interception for ERSOW Middle Size League Soccer Robot,” in *2025 International Electronics Symposium (IES)*, IEEE, Aug. 2025, pp. 383–389. doi: 10.1109/IES67184.2025.11161124.

[15] S. Yano, T. Nakashima, Y. Kusunoki, and H. Akiyama, “RoboCup Soccer Ball and player trajectory prediction using LSTM,” in *The 39th Annual Conference of the Japanese Society for Artificial Intelligence*, Japanese Society for Artificial Intelligence, 2025, ch. 2S1-GS-2-01, pp. 1–2. doi: 10.11517/pjsai.JSAI2025.0_2S1GS201.

[16] K. T. Putra, W. Muldayani, Sumardi, Moch. S. Komar, Z. Albatati, and F. Mujaahid, “A Robust Visual-IMU-Wheel Odometry Using PID Controller for Autonomous Soccer Robots,” in *2022 International Conference on Electrical Engineering, Computer and Information Technology (ICEECIT)*, IEEE, Nov. 2022, pp. 76–81. doi: 10.1109/ICEECIT55908.2022.10030341.

[17] C. Premachandra and M. Tamaki, “A Hybrid Camera System for High-Resolutionization of Target Objects in Omnidirectional Images,” *IEEE Sens. J.*, vol. 21, no. 9, pp. 10752–10760, May 2021, doi: 10.1109/JSEN.2021.3059102.

[18] Rizka Rizka, S. Nasution, F. Aulia, and Supiyandi Supiyandi, “Penerapan Metode Segmentasi Warna HSV untuk Deteksi Objek Berbasis Warna pada Citra Digital,” *Router : Jurnal Teknik Informatika dan Terapan*, vol. 3, no. 4, pp. 11–24, Dec. 2025, doi: 10.62951/router.v3i4.706.

[19] K. Joshi and B. Patil, “Prediction of Surface Roughness by Machine Vision using Principal Components based Regression Analysis,” *Procedia Comput. Sci.*, vol. 167, pp. 382–391, 2020, doi: 10.1016/j.procs.2020.03.242.

[20] N. Andriyanov, “Estimating Object Coordinates Using Convolutional Neural Networks and Intel Real Sense D415/D455 Depth Maps,” in *2022 8th International Conference on Information Technology and Nanotechnology, ITNT 2022*, Institute of Electrical and Electronics Engineers Inc., 2022. doi: 10.1109/ITNT55410.2022.9848700.

[21] O. Aribisala, “Data Quality Assessment and Preprocessing Techniques for Enhancing Machine Learning Model Performance.,” *International Journal of Innovative Research in Engineering & Multidisciplinary Physical Sciences*, vol. 12, no. 3, Jun. 2024, doi: 10.37082/IJIRMPS.v12.i3.232799.